



Catalytic stepwise pyrolysis of packaging plastic waste

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ABSTRACT

In this paper the combination of catalytic and stepwise pyrolysis is explored. A mixture of polyethylene (PE), polypropylene (PP), polystyrene (PS), poly(ethylene terephthalate) (PET) and poly(vinyl chloride) (PVC), which resembles real municipal plastic waste, has been pyrolysed in a 3.5 dm³ semi-batch reactor at 440 °C for 30 min using a ZSM-5 zeolite as catalyst. A low temperature (300 °C) dechlorination step has been carried out both with and without catalyst. It has been proved that the application of such dechlorination step gives rise to a 75 wt% reduction of chlorine in the liquid fraction. However, such step has a negative influence on the catalyst, which loses some catalytic activity. The optimum procedure in terms of quality and chlorine content of the products is the combination of first a low temperature step without catalyst, and second the catalytic pyrolysis step.

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1. Introduction

PVC is one of the most widely used polymers. Its polar nature, provided by the chlorine atom in its repeat unit, enables to combine this polymer with a much wider range of additives than any other polymer, yielding a wide range of plastic compounds with very different properties. Consequently, it can be used in a significant number of applications, ranking second in total thermoplastic worldwide production, just following polyolefins [1]. However, the presence of chlorine in the PVC backbone is also the key issue when it comes to waste PVC management, due to the chlorinated compounds which can arise during waste treatment operations as a consequence of the molecular instability of PVC towards heat and light [2].

Nowadays PVC is mainly used in the construction sector, followed by electronic equipment and automotive applications. Although the use of this material for packaging applications has decreased during the last years, 4% of total packaging goods was estimated to be made of PVC in 2007, including thin-wall containers, small bottles, film, sheets and lids [3]. Since nowadays PVC is a minority product in packaging waste, it is not specifically recovered from such waste for recycling, as it is in the case of the construction PVC wastes [4]. For this reason the presence of PVC in rejected packaging waste streams is still very common [5]. These rejected streams are the fractions of packaging waste which cannot be mechanically recycled since they are composed of different

and much intermingled plastic materials, and consequently, they are nowadays incinerated or landfilled.

Pyrolysis is a process which has received renewed attention in the last years for recycling plastic waste, converting them in potentially useful liquids and gases [5–7]. In the pyrolysis process (heating in an oxygen free atmosphere), the organic components of the material are decomposed generating liquid and gaseous products while the inorganic ingredients (fillers, metals, etc.) remain practically unaltered and free of the binding organic matter, allowing their separation to be recovered and reused. Therefore, pyrolysis is an attractive and alternative recycling technique for rejected plastics waste streams, provided that the formation of chlorinated compounds is prevented, since they are very detrimental for using pyrolysis products as fuels or chemical reactants.

The pyrolysis process can be improved by means of dechlorination methods which prevent the generation of chlorinated compounds when PVC containing samples are pyrolysed. The dechlorination methods more frequently reported in the literature are stepwise pyrolysis and pyrolysis with adsorbents. In stepwise pyrolysis, a low temperature step is carried out in order to remove chlorine from the original sample as HCl, since it is a well-known fact that PVC decomposes between 250 and 320 °C, depending on the material and the stabilizers and additives [8,9]; in a second step the sample is pyrolysed as in a conventional one-step run. On the other hand, the addition of adsorbents to the sample reduces the HCl emission since the evolved HCl is trapped by means of physical and/or chemical adsorption and is thus retained in the solid fraction.

There is plenty of information in the literature about dechlorination in the thermal decomposition of PVC either alone or mixed

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Table 1

Origin, application and proportions of the plastics used in the mixture.

Plastic name	Acronyms	Origin	State	Application	Proportion (wt%)
Polyethylene	PE	Repsol Química	Virgin	Household	40
Polypropylene	PP	Repsol Química	Virgin	General	35
Polystyrene	PS	Dow Chemical	Virgin	General	18
Poly(ethylene terephthalate)	PET	Remaplast S.A. ^a	Recycled	Bottles	4
Poly(vinyl chloride)	PVC	Gaiker ^b	Recycled	Bottles	3

^a Spanish Company devoted to municipal plastics recycling.^b Spanish Technology Centre dedicated to research and innovation.

with other plastics [e.g. 10–15]; on the other hand, zeolites have been reported to be effective for improving the quality of polymeric materials pyrolysis products, as well as to enable to operate at milder pyrolysis conditions (lower temperatures and shorter reaction times) than in one-step conventional thermal pyrolysis, mainly due to its appropriate textural properties and high acidity [e.g. 7,16–18]. Due to this high acidity, it might be expected that zeolites are somewhat resistant to HCl.

On the contrary, it is not easy to find information in the literature about the combined effects of both the dechlorination step and the catalyst together in the pyrolysis process. Most of the catalytic dechlorination studies of PVC mixtures carried out up to now have focused on using the catalysts as adsorbents for the evolved HCl [19,20] or as HCl formation inhibitors [21–23]. To the best of our knowledge, only two papers, published by Zhou et al. [24] and Yanik et al. [25], have shown the combined effect of the low temperature dechlorination step and the catalytic action, but using simple binary mixtures (PP/PVC, PE/PVC, PS/PVC) and with catalysts different from zeolites.

In a previous paper of the authors [9] stepwise pyrolysis experiments with the very same sample as that used in this paper but without catalyst were reported. The results obtained showed that a reduction of more than 50 wt% of chlorine content in the liquid fraction could be achieved if stepwise pyrolysis was used. On the other hand, the authors have also reported [26,27] that the use of ZSM-5 zeolite catalyst enhances the quality of pyrolysis products. The partial objectives of this paper are to study if the dechlorination step is also effective when the zeolite is used and to elucidate if the dechlorination step could affect the catalyst activity, bearing in mind that the overall objective of this study is to obtain high quality pyrolysis liquids with low chlorine content. The experiments have been carried out with a complex plastic mixture which resembles real packaging waste streams. An exhaustive characterization of all the pyrolysis products obtained has been included in the paper, in order to provide useful information for potential industrial applications.

2. Materials and methods

2.1. Raw materials and catalyst

A mixture of plastics which simulates the composition of real plastic waste streams rejected from industrial sorting plants was prepared for the pyrolysis experiments. The origin, application and proportions of each plastic in the mixture are presented in Table 1. The composition of the sample was established based on the composition of real samples rejected from an industrial plant located in the north of Spain (Amorebieta), where the packaging waste of the province of Biscay is sorted [5]. All the plastic materials were used in pellet size (≈ 3 mm) for the pyrolysis experiments. Additionally finely ground samples (≤ 1 mm) were prepared for characterization purposes; the composition and characteristics of the sample used for the pyrolysis experiments are shown in Table 2.

The catalyst used for the pyrolysis experiments was a commercial ZSM-5 zeolite provided by Zeolist International; in all the experiments it was used as received and in a plastic/catalyst

Table 2Moisture, ash and elemental composition (wt%) and HHV (MJ kg⁻¹) of the plastic mixture.

Parameter	Moisture	Ash	C	H	N	Cl	Others ^a	H/C ratio	HHV
Value	0.1	0.0	84.7	12.5	<0.1	1.1	1.5	1.8	43.9

^a By difference.

mass ratio of 10/1. Its main characteristics are presented in Table 3. The textural properties of the catalyst were determined by means of nitrogen adsorption–desorption isotherms at 77 K in an AUTOSORB-1 Quantachrome equipment. Surface areas were calculated by means of BET equation and external surface areas were obtained by applying the *t*-plot method. Total pore volume was measured at $P/P_0 = 0.99$. The acidity of the catalyst was measured by temperature-programmed desorption (TPD) of ammonia with a Micromeritics AutoChem 2910 instrument.

2.2. Experimental

The pyrolysis experiments were carried out using an unstirred stainless steel 3.5 dm³ reactor (cylindrical, 0.15 m diameter) in semi-batch operation. In all the experiments, 100 g of plastic sample was placed into the reactor and nitrogen was passed through at a rate of 1 dm³ min⁻¹ through the whole experiment; nitrogen was feed at the bottom of the reactor under a diffusion plate placed inside the reaction chamber, also at the bottom, which distributed nitrogen all around the reaction chamber. The vapours leaving the reactor flowed to a series of running water cooled gas-liquid separators where the condensed liquids were collected. The uncondensed products were passed through an activated carbon column and collected as a whole in Tedlar plastic bags. The experimental set-up used in the experiments is presented in Fig. 1.

Three different pyrolysis methods were explored. (1) “Conventional catalytic pyrolysis”, in which the plastic sample was mixed with the catalyst and the system was heated at a rate of 20 °C min⁻¹ to 440 °C, and maintained there for 30 min. (2) “Catalytic stepwise pyrolysis”, in which the plastic sample was mixed with the catalyst, a dechlorination step was carried out at 300 °C during 60 min and then the temperature was raised at 20 °C min⁻¹ to 440 °C to complete the pyrolysis process. (3) “Non catalytic dechlorination step + catalytic pyrolysis”, in which the dechlorination step was applied to the plastic sample alone (without catalyst) and then the catalyst was added, carrying out the final pyrolysis as in the other methods. In this last method, it was necessary to cool down the system with the plastic sample to room temperature after the dechlorination step, extract the dechlorinated melted sample,

Table 3

Textural and acid properties of the ZSM-5 zeolite.

BET surface area (m ² g ⁻¹)	412.0
External surface area (ESA) (m ² g ⁻¹)	65.88
Micropore volume (MPV) (cm ³ g ⁻¹)	0.100
Total pore volume (TPV) (cm ³ g ⁻¹)	0.397
Total acidity (mmol NH ₃ g ⁻¹)	0.176

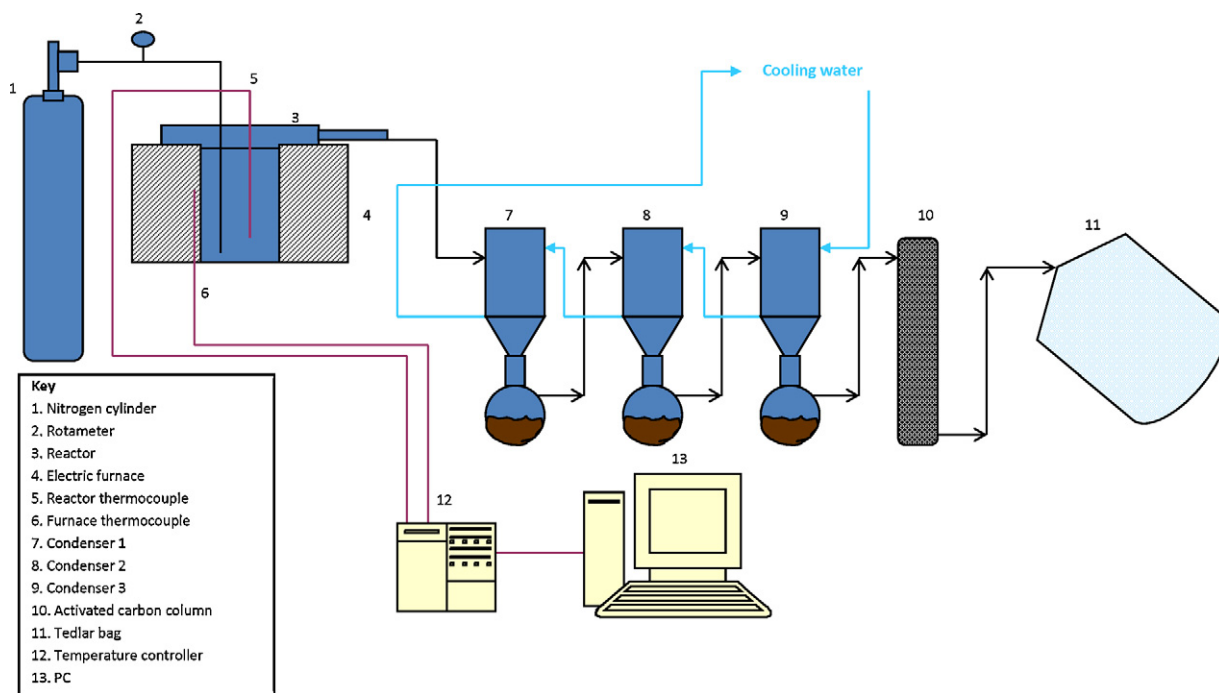


Fig. 1. Flow sheet of the experimental set-up used.

freeze it in liquid nitrogen, grind it in a mill, mix it with the catalyst and place them together again in the reactor, to carry out the catalytic second step. The operational conditions selected for the dechlorination step were 300 °C and 30 min since with such conditions the best dechlorination rates were achieved in the previous paper concerning plastic waste dechlorination published by the authors [9].

Additionally, the results obtained in a “conventional one-step thermal pyrolysis” (which consisted in heating the plastic sample without catalyst at a rate of 20 °C min⁻¹ to 440 °C followed by a dwell of 30 min) have been included in Tables 6 and 10 in order to satisfactorily explain the results obtained in the three pyrolysis methods explored in this paper.

The amount of solids remaining in the reactor and the liquids condensed in the gas-liquid separators were weighed after pyrolysis and the pyrolysis yields were calculated as weight percentage with respect to the raw material pyrolysed. The amount of input zeolite was subtracted from the amount of total solids remaining in the reactor and the solid yield was calculated with respect to input plastic material (without catalyst). Gas yields were normally determined by difference; in some experiments specifically devoted to directly quantify the amount of gases, a closure of the mass balance of about 90 wt% was obtained.

2.3. Analytical techniques

Both the raw material and the solid and liquid products obtained were characterized using the following techniques. The moisture and ash contents of the samples were determined by thermogravimetric analysis according to D3173-85 and D3174-82 ASTM standards respectively, and the elemental composition with an automatic CHN analyser. Method 5050 of the Environmental Protection Agency (EPA) of the United States was used for chlorine determination. The higher heating value (HHV) was determined with an automatic calorimetric bomb.

Additionally, pyrolysis liquids were also analysed by gas chromatography coupled with a mass spectrometry detector (GC–MS), Agilent 6890 and Agilent 5973 respectively. The characteristics of

Table 4

GC–MS equipment and method characteristics.

Equipment	Agilent 6890 (GC) & Agilent 5973 (MS)
Column	HP5MS
Carrier gas	He
Carrier gas flow	1.0 mL/min
Initial temperature/initial time	40 °C/10 min
Heating rate	8 °C/min
Final temperature/final time	280 °C/10 min
Injection temperature	280 °C
Injection volume	1.0 µL (split 100:1)
MS detector temperature (quad/source)	150 °C/230 °C
Acquisition mode	Scan
Mass range	50–550 a.m.u.

the method used are shown in Table 4. Identification of the compounds was based on comparison of the retention times with those of calibration samples, on computer matching against commercial library of mass spectra (Wiley7n) and also comparing to mass spectra literature data. The chromatograph was equipped with a non-polar column which is appropriate for a general purpose analysis of hydrocarbons mixtures. When the match quality of the identification result provided by the MS search engine was lower than 85% of spectrum coincidence, the result was not considered

Table 5

GC–FID/TCD equipment and method characteristics.

Equipment	Agilent 7890
Columns	HP-Molesieve HP-PLOT/Q
Carrier gas	He
Carrier gas total flow	255 mL/min
Injection temperature	150 °C
Injection volume	25 µL (split variable)
Initial temperature/initial time	40 °C for 9.5 min 20 °C/min to 100 °C for 2.5 min
Oven temperature program	10 °C/min to 180 °C for 0 min 20 °C/min to 200 °C for 4 min

Table 6
Pyrolysis yields (wt%).

Method	Liquids	Gases	Solids
Conventional thermal pyrolysis	79.3 ± 1.9	17.7 ± 1.9	3.0 ± 0.3
Conventional catalytic pyrolysis	56.9 ± 3.0	40.4 ± 3.0	3.2 ± 0.2
Catalytic stepwise pyrolysis	69.0 ± 0.1	29.0 ± 0.1	2.0 ± 0.0
Non-catalytic dechlorination step + catalytic pyrolysis	56.8 ± 2.0	41.2 ± 2.0	2.0 ± 0.2

valid and these compounds are classified as “Not identified” in this paper.

Concerning pyrolysis gases, they were analysed by means of a gas chromatograph (Agilent 7890) coupled with a thermal conductivity and a flame ionization detector (GC-TCD/FID). Table 5 shows the characteristics of the method used. The HHV of the gases was theoretically calculated according to their composition and to the HHV of the individual components.

3. Results and discussion

3.1. Pyrolysis yields

The liquid, gas and solid yields (wt%), obtained with the three mentioned pyrolysis methods and those of the conventional one-step thermal run are presented in Table 6. If the conventional (one-step) thermal and conventional (one-step) catalytic pyrolysis experiments are compared, it can be stated that the catalyst plays an important role in pyrolysis products distribution, producing more gases and fewer liquids than in the thermal run. A more thorough discussion about the effect of ZSM-5 zeolite in pyrolysis of plastic waste has been presented elsewhere [26,27]. When the dechlorination step is carried out in the presence of the catalyst, a somehow intermediate behaviour in pyrolysis yields can be observed, producing an increase in liquid yield and a decrease in gas yield compared with the conventional catalytic experiment, which indicates that the zeolite loses some activity during the dechlorination step.

In the previous paper published by the authors [9], the effect of the dechlorination step in the absence of catalyst was just the contrary. In the stepwise non-catalytic run more gases and less liquid than in the one-step non-catalytic run were obtained. This was attributed to some partial thermal cracking of the polymers and rearrangement of the chemical structures that may have taken place in the dechlorination step, which facilitates the production of gas in the pyrolysis stage. The fact that in the present paper such effect (gas yield increase when the dechlorination step is applied) has not been observed may be explained as follows: (1) in one-step

pyrolysis the effect of the catalyst is to increase gas yield and decrease liquid yield, (2) in stepwise catalytic pyrolysis, although on the one hand gas yield should increase due to the effect of the dechlorination step, on the other hand it should decrease due to the loss of activity of the catalyst, and (3) this later decrease seems to be greater than the increase due to the dechlorination step.

This partial cracking of the polymers together with the fact that the plastic sample is melted during the dechlorination step could be the reason why the catalyst activity is lost in the actual work, since the cracked and melted fragments may physically block the catalyst pores during the dechlorination step. Consequently, the catalytic effect of the zeolite in the subsequent pyrolysis step could be hindered. In addition, Table 6 shows that when the catalyst is added to the sample after the dechlorination step, almost the same yields as in the conventional catalytic experiment are obtained, which confirms that the dechlorination step had brought about the loss of activity of the zeolite.

Table 6 shows that in every case, a small quantity (2.0–3.2 wt%) of solid residue was obtained. The solid residues were a mixture of unconverted plastic and char, a carbonized product formed during the pyrolysis process. The formation of char in pyrolysis of polymeric materials is a well documented fact which has been reported before by the authors [28,29] as well as by other research groups [30,31]. The solid residue in the conventional catalytic run is somewhat higher than in the other runs, which may indicate that the dechlorination step somewhat prevents the formation of char.

As far as pyrolysis yields are concerned, the results indicate that a staged process (PVC dechlorination before the catalytic pyrolysis stage) could be the best option, but before putting forward such conclusion the pyrolysis products characteristics must be studied.

3.2. Characteristics of the pyrolysis liquids

The results obtained in the GC–MS analyses of the pyrolysis liquids are presented in Tables 7 and 8. These results have been reported as % area calculated with respect to total ion, in order to be more representative of absolute abundance.

Table 7 shows all the compounds identified by GC–MS grouped in three categories according to their carbon atom number: C5–C9, C10–C13 and >C13. Additionally, the proportion of aromatics and non aromatics of each of these fractions has also been included in this table, as well as total aromatics, non aromatics and unidentified compounds. As it can be seen in Table 7, C5–C9 is the main fraction of the pyrolysis liquids in a conventional catalytic experiment, amounting to more than 80% area. The formation of light hydrocarbons in the pyrolysis of polyolefinic plastic waste in presence of ZSM-5 zeolite has been reported before [32,33]; Serrano

Table 7
Fractions of interest in pyrolysis liquids (% area).

Fractions	Conventional catalytic pyrolysis	Catalytic stepwise pyrolysis	Non-catalytic dechlorination step + catalytic pyrolysis
C5–C9	Aromatics	78.7	69.6
	Non aromatics	2.8	4.8
	Total	81.5	74.4
C10–C13	Aromatics	9.5	1.5
	Non aromatics	0.0	0.0
	Total	9.5	1.5
>C13	Aromatics	6.9	9.5
	Non aromatics	0.0	0.0
	Total	6.9	9.5
Total aromatics		95.1	80.6
Total non aromatics (branched olefins)		2.8	4.8
Not identified		2.1	14.6

Table 8

GC–MS analysis of the pyrolysis liquids (% area).

Molecular formula	Assignment	Conventional catalytic pyrolysis	Catalytic stepwise pyrolysis	Non-catalytic dechlorination step + catalytic pyrolysis
C ₅ H ₁₀	Methyl butene	1.1	0.9	1.2
C ₇ H ₈	Toluene	12.3	8.5	13.1
C ₉ H ₁₈	Dimethyl heptene	1.7	3.9	2.1
C ₈ H ₁₀	Ethyl benzene	10.6	8.0	12.3
C ₈ H ₁₀	Xylenes	10.1	2.9	7.4
C ₈ H ₈	Styrene	31.4	44.9	31.5
C ₉ H ₁₂	Propyl benzene	1.2	n.d. ^a	1.0
C ₉ H ₁₂	Ethyl methyl benzenes	3.7	n.d. ^a	2.2
C ₉ H ₁₀	Methyl styrene	4.5	4.4	6.4
C ₉ H ₁₂	Trimethyl benzene	2.0	n.d. ^a	1.6
C ₉ H ₁₀	Propenyl benzene	1.1	0.9	1.5
C ₉ H ₈	Indene	1.9	n.d. ^a	1.6
C ₁₀ H ₁₀	Methyl indenenes	3.3	0.7	2.8
C ₁₀ H ₈	Naphthalene	2.3	n.d. ^a	1.6
C ₁₁ H ₁₀	Methyl naphthalene	3.3	0.8	2.2
C ₁₂ H ₁₂	Ethyl naphthalene	0.7	n.d. ^a	n.d. ^a
C ₁₅ H ₁₆	Phenyl propyl benzene	1.3	2.1	1.9
C ₁₅ H ₁₄	Phenyl cyclopropyl benzenes	2.0	n.d. ^a	2.6
C ₁₅ H ₁₄	Phenyl propenyl benzene	n.d. ^a	1.5	n.d. ^a
C ₁₆ H ₁₆	Phenyl butenyl benzene	0.9	3.6	0.9
C ₁₆ H ₁₂	Phenyl naphthalenes	1.7	0.9	1.9
C ₁₆ H ₁₄	Dimethyl phenanthrene	1.1	1.4	1.5
Total identified		97.9	85.4	97.4

^a Not detected.

et al. [33] suggested that in the pyrolysis of polyolefins this catalyst leads the reaction through an end-chain scission pathway, yielding light hydrocarbons as primary products, instead of the typical polyolefinic random scission pathway which takes place in thermal pyrolysis.

When the dechlorination step is carried out in the presence of the zeolite, the percentage of light (C5–C9) and medium (C10–C13) hydrocarbons significantly decrease (to 74.4 and 1.5% area respectively), and the yield of heavy hydrocarbons increases up to 9.5% area. This fact is in agreement with the previously mentioned loss of activity of the catalyst that is produced during the dechlorination step. In fact, when the catalyst is added after the dechlorination step, the C5–C9 yield is quite similar to that of the conventional catalytic run (81.5 and 82% area), which indicates that the catalyst has worked. However, there are differences in the C10–C13 and >C13 yields, which are lower and higher respectively than in the conventional catalytic run. The very same occurred in the previous study of the authors [9], when stepwise pyrolysis was compared to conventional pyrolysis but both without using catalyst. It was then suggested that some degradation and rearrangement of polymer structures take place during the dechlorination step, leading to different pyrolysis pathways and consequently to different pyrolysis products.

Whenever the dechlorination step is carried out, higher >C13 yield is obtained. It seems plausible that such incipient degradation enables larger molecules to evolve from the reactor in the subsequent pyrolysis step. This reasoning may explain also the fact previously mentioned that less solid residue is obtained when the dechlorination step is carried out. In the case of stepwise catalytic pyrolysis there is a combination of the catalytic effect (less liquid, more gas) and the dechlorination step effect (more >C13 hydrocarbons in liquids) taking place at the same time, which justifies the contradictory fact that less liquid, more gas but more heavy liquid hydrocarbons (>C13) are produced.

Concerning aromatics and non aromatics content, Table 7 shows that the pyrolysis liquids derived from ZSM-5 conventional catalytic pyrolysis have a highly aromatic nature (>95% area). This seems at first sight quite surprising since the plastic sample used in this work is mainly composed of polyolefins; however, there is evidence in the literature that high contents of aromatics can

be obtained in pyrolysis of polyolefins, though there is no agreement about the aromatics formation mechanisms and two main routes have been suggested [8]. Diels–Alder reactions followed by dehydrogenation and unimolecular cyclation reactions followed by dehydrogenation, which some authors call “pyrosynthesis”.

When zeolite-type catalysts are used, the aromatization reactions are much stronger. Many other authors [e.g. 17,34], as well as the authors in previous works [26,27], have also obtained liquids with high level of aromatics when ZSM-5 zeolite is used, which is attributed to the high number of Brönsted acid sites contained within this zeolite, which promote aromatization reactions. This high proportion of aromatics is an interesting characteristic for pyrolysis liquids because with a point of view of their using in refinery processes, a pure aromatic liquid fraction is more valuable than a mixture of aromatics and olefins [8]. Therefore, it can be stated that ZSM-5 catalyst is a good selection for plastic pyrolysis processes.

It can be seen in Table 7 that when the dechlorination step is carried out with the catalyst mixed with the sample, the production of aromatics is lower than in conventional catalytic pyrolysis. However, when the catalyst is added after the dechlorination step the aromatics content is almost as high as in the conventional catalytic run. It is also remarkable that with the application of the catalytic dechlorination step produces higher “not identified” compounds than in the conventional pyrolysis run are yielded (14.6 vs. 2.1% area). Quite similar results were obtained by the authors in their previous paper about dechlorination in thermal non-catalytic pyrolysis [9], in which the yield of “not identified” compounds in the non-catalytic stepwise pyrolysis was also higher than 10% area.

Taking into account that the match quality of the identification provided by the MS search engine is lower for larger molecules, this high proportion of unidentified compounds may be because somewhat larger molecules evolve from the reactor due to the partial decomposition and rearrangement of the polymer structures that take place when the dechlorination step is used, as it has been mentioned before.

Table 7 also shows that there is a relationship between the different carbon atom number fractions and the aromatics content. As it can be seen, all the compounds with carbon atom number greater than C10 are aromatic compounds, therefore all the non

Table 9
Elemental composition (wt%) and HHV (MJ kg⁻¹) of the pyrolysis liquids.

Experiment	Conventional catalytic pyrolysis	Catalytic stepwise pyrolysis	Non-catalytic dechlorination step + catalytic pyrolysis
C	87.6	87.7	83.9
H	11.2	11.9	11.9
Cl	1.2	0.3	0.3
Others ^a	0.0	0.1	3.9
H/C Ratio	1.5	1.6	1.7
HHV	42.2	44.4	43.3

^a By difference.

aromatic compounds are contained in the C5–C9 fraction compounds, and more specifically they are C5–C9 branched olefins, as will be shown in Table 8. Taking into account the nature of the sample used (mainly composed of polyolefins) and that pyrolysis reactions require significant amounts of available hydrogen to form stable compounds, the following global reaction mechanism may be suggested: (1) polymers cracking to yield olefins and olefinic radicals and (2) aromatics formation from olefinic structures, either radicals or stable structures, since the olefinic stable compounds formed in the process can further react and form aromatics if the reaction conditions are favourable to bring it about, i.e., at high temperatures and long residence times.

The main identified components of the pyrolysis liquids are presented in Table 8. As it can be seen, pyrolysis liquids are composed of hydrocarbons from C5 to C16. The main chemical families are the following: branched olefins, one ring-aromatics, indenenes, two ring-aromatics (naphthalenes and phenyl benzenes) and three ring-aromatics (phenyl naphthalenes and phenanthrenes).

It can be seen that once more, the liquids derived from the conventional catalytic pyrolysis and the non-catalytic dechlorination step + catalytic pyrolysis experiments produced quite similar results. Both yielded more than 30% area of styrene and around 10% area of toluene, ethyl benzene and xylenes, which are the four main compounds of the liquid fraction derived from both experiments.

The liquids of the catalytic stepwise pyrolysis experiment are somewhat different; they contain much more styrene (≈45% area) and less of the other compounds. Serrano et al. [35] reported that in the acid catalyzed cracking of PS, different products such as ethyl benzene or methyl benzenes can be formed instead of styrene due to the carbenium nature of the acid catalyzed cracking, while when no catalyst is used thermal cracking leads to the formation of styrene and its corresponding dimers and trimers as the main products. In accordance with Serrano, the higher content of styrene and of other monoaromatics in the stepwise catalytic experiment may be due to the fact that the catalyst has lost its activity during the dechlorination step.

In fact, the sum of the area percentages of toluene, ethyl benzene, xylenes and styrene in the three experiments is quite similar (64.4, 64.3 and 64.3% area). So when the catalyst is not effective (as in the catalytic stepwise experiment), the formation of styrene is favoured to the detriment of the other three compounds.

Another significant difference in the pyrolysis liquids derived from the catalytic stepwise pyrolysis compared to those obtained in the other two experiments, is that the liquids from the former experiment have lower quantities of indenenes and naphthalenes. This difference may be again attributed to the loss of activity of the catalyst; however, due to the complexity of the reactions and products involved, an explanation to justify this difference is yet to be advanced. It must be borne in mind that there is a great difficulty in discussing polymer pyrolysis reaction mechanisms, especially when such a complex system as that used in this study is considered (five polymeric input materials and innumerable decomposition products), and even more in the presence of catalyst and in a multi-step process. On the one hand, when catalyst is used in the pyrolysis

process two broad types of decomposition mechanisms take place simultaneously: thermal cracking, which in turn can follow different mechanisms (random chain scission, end-chain scission and/or group side elimination), and catalytic cracking (carbenium ions adsorption in the catalyst surface, beta scissions and desorption). As a result a wide variety of products are generated, which in turn will react with each other, resulting in a countless number of possible reaction mechanisms. On the other hand, when a first low temperature step is used, there is an arrangement of the polymer structures which influences the reactions of the subsequent pyrolysis process, making it further complicated to explain the reaction mechanisms.

To end with the results concerning the GC–MS analysis, it is worth noting that there are not chlorinated species among the compounds reported in Table 8, most probably because the proportion of chlorine in the liquids is quite low and is distributed among several chlorinated species. As a consequence, the chlorinated species are present in a concentration below the detection limit established in the GC–MS method.

The elemental composition and higher heating value of the pyrolysis liquids obtained in each experiment are presented in Table 9. It can be seen that there are no significant differences among the elemental compositions of the liquids, which are mainly composed of carbon (84–88 wt%) and hydrogen (9–10 wt%). It is also worth noting that surprisingly the carbon content of the liquid obtained in the non-catalytic dechlorination step + catalytic pyrolysis experiment was lower than that obtained in the other two experiments, and consequently the H/C ratio is higher. This seems at first sight inconsistent with the aromatics contents reported in Table 7, since the liquids of this experiment have similar contents of aromatics as the conventional catalytic experiment and therefore similar H/C ratios were expected. However it must be born in mind that the relationship between aromatics content and H/C ratio is not univocal since aromatics have higher H/C ratios as more abundant and longer are the aliphatic chains linked to the aromatic rings.

It is worth mentioning that in all the cases pyrolysis oils have very high HHV, similar to those of conventional liquid fuels, so they may be considered as an appropriate alternative to fossil fuels. Obviously, the presence of chlorine in the liquids is rather detrimental for this application. A thorough discussion concerning chlorine content and its evolution will be commented in Section 3.5.

3.3. Gas composition

The composition of the pyrolysis gases is presented in Table 10. It has been calculated in a free chlorine basis, since in a potential industrial process the evolved HCl should be separated before the application of the gases. The chlorine content of the gases will also be discussed in Section 3.5. Table 10 shows that pyrolysis gases are mainly composed of hydrocarbons ranging from C1 to C6, together with some hydrogen and some carbon dioxide. The hydrocarbons have been grouped in three fractions (C1–C2, C3–C4 and C5–C6) in

Table 10GC-TCD/FID analysis (wt%) and HHV (MJ kg⁻¹) of pyrolysis gases.

Experiment	Conventional catalytic pyrolysis	Catalytic stepwise pyrolysis	Non-catalytic dechlorination step + catalytic pyrolysis
H ₂	0.8	0.6	0.7
CO	0.7	0.3	0.4
CO ₂	4.9	2.9	2.6
C1–C2	24.6	10.9	31.6
C3–C4	57.0	60.4	61.7
C5–C6	11.9	25.0	3.0
HHV	46.0	48.4	49.4

order to better analyze the effect of the different pyrolysis methods used. C3–C4 hydrocarbons are the main fraction and have comparable yields in all the experiments, which is in agreement with the results of other authors who have also reported that C3 and C4 are the main fractions of the gases of pyrolysis of polyolefins when ZSM-5 zeolite is used [36,37].

Concerning the influence of the different dechlorination methods in gas composition, Table 10 shows that the lowest proportions of light hydrocarbons (C1–C2) and the highest of heavy hydrocarbons (C5–C6) were obtained in the stepwise catalytic pyrolysis run (10.9 wt% and 25 wt% respectively). This is quite in accordance with the previous results concerning the characteristics of the liquids, and it is attributed to the fact that the zeolite loses its activity during the dechlorination step. When the catalyst is added after such step (non-catalytic dechlorination step + catalytic pyrolysis), the highest C1–C2 fraction proportion (31.6 wt%) and the lowest C5–C6 hydrocarbons proportion (3.0 wt%) is obtained, therefore even stronger cracking than in the conventional catalytic process takes place, so the dechlorination step somehow facilitates the subsequent cracking action of the catalyst.

It has to be mentioned that in the non-catalytic dechlorination step + catalytic pyrolysis experiment, more than 90 wt% of pyrolysis gas is composed of C1–C4 hydrocarbons, which confers this gas fraction a remarkable HHV (49.4 MJ kg⁻¹). Anyway, the gases obtained in any of the pyrolysis runs have HHVs comparable to that of natural gas (48–53 MJ kg⁻¹), therefore pyrolysis gases have a great potential for power generation. It has to be mentioned that the calculation of the HHV of the gases has been done in HCl free basis since, as it has been mentioned before, in a potential industrial pyrolysis process the evolved HCl should be removed before the utilization of the gases.

3.4. Pyrolysis solids

The composition of the pyrolysis solids is presented in Table 11. In all the cases the solids are mixed with the catalyst and this is the reason for the high “others” content. The carbon and hydrogen contents of the solids of the experiments carried out with dechlorination step are lower than when such step is not performed, which could be in relation with the slightly lower solid yields obtained in these experiments (shown in Table 6), and which could indicate that the dechlorination step favours the completion of the

Table 11Moisture, elemental composition (wt%) and HHV (MJ kg⁻¹) of the pyrolysis solids.

Experiment	Conventional catalytic pyrolysis	Catalytic stepwise pyrolysis	Non-catalytic dechlorination step + catalytic pyrolysis
Moisture	0.7	1.0	0.9
C	23.0	14.4	13.8
H	2.2	0.9	1.0
Cl	0.4	0.4	0.4
Others ^a	73.7	83.3	83.9
H/C ratio	1.1	0.8	0.9
HHV	9.4	4.9	6.3

^a By difference.**Table 12**

Chlorine in the pyrolysis fractions (wt%).

Method	Cl in liquids	Cl in gases ^a	Cl in solids
Conventional thermal pyrolysis	0.2	5.3	<0.1
Conventional catalytic pyrolysis	1.2	1.0	0.4
Catalytic stepwise pyrolysis	0.3	3.0	0.4
Non-catalytic dechlorination step + catalytic pyrolysis	0.3	2.2	0.4

^a Calculated by difference taking into account that there is 1.1 wt% chlorine in the original sample.

pyrolysis process which is not completed in the conventional catalytic experiment. As it has been mentioned for the other pyrolysis products, the discussion concerning chlorine content and distribution will be included in Section 3.5.

3.5. Distribution of chlorine in the pyrolysis products

Table 12 shows the percentage of chlorine in the products obtained after each experiment. In order to compare the chlorine proportion between the liquids obtained in a conventional thermal experiment and those obtained in a conventional catalytic experiment, the chlorine content of the liquids obtained in the conventional thermal pyrolysis run at 440 °C has been included in Table 12. It can be seen that the chlorine content of the liquids after a conventional catalytic pyrolysis run is 1.2 wt%, which is higher than in the conventional thermal run (0.2 wt%) and it may condition the potential application of such liquids. Yanik et al. [25] studied the chlorine content of oils derived from catalytic pyrolysis with different types of catalysts and did obtain the highest concentration of chlorine in pyrolysis oils after using a SiO₂–Al₂O₃ nature catalyst, as is the one used in this work. Yanik et al. attributed this fact to a certain retention of HCl by the solid catalyst; this enables the polymer derived products to react with HCl, thereby generating chlorinated compounds. Therefore, in this paper the effect of the catalyst concerning chlorine content in liquids follows the same tendency as in the case of Yanik. This result indicates that the quality of pyrolysis liquids, in terms of chlorine content, is lower in catalytic pyrolysis

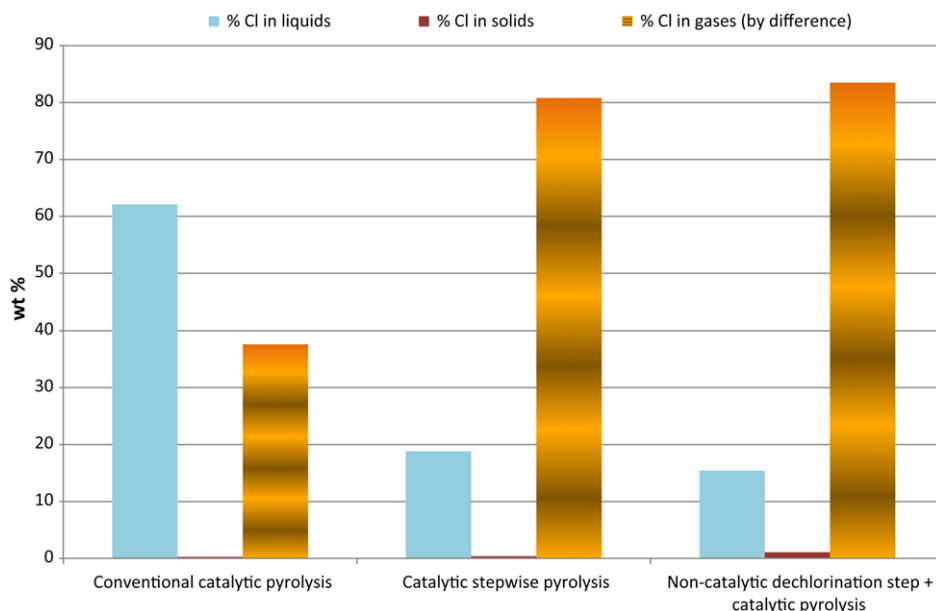


Fig. 2. Chlorine distribution among the pyrolysis fractions (wt%).

than in thermal pyrolysis. For this reason, the dechlorination step is an essential stage in catalytic pyrolysis of PVC containing plastic waste.

When the dechlorination step is applied either with or without catalyst, the chlorine content of the liquids decreases from 1.2 to 0.3 wt%, which is equivalent to a 75% reduction of chlorine in the liquids with respect to the conventional catalytic pyrolysis liquids. Since chlorine evolves as HCl (mainly generated in the first step), Table 10 shows higher contents of chlorine in the gases when the dechlorination step is used (2.2–3.0 wt%) than that obtained in conventional catalytic pyrolysis (1.0 wt%). Nevertheless, as it has been mentioned before, the objective of this step is precisely to separate the chlorine as HCl, since in an industrial process HCl could be absorbed in alkaline solutions by means of wet scrubbers and the gases generated in the subsequent pyrolysis step would be free of chlorine.

The effect of the dechlorination step can be more clearly analyzed in Fig. 2, which shows the chlorine weight % distribution among the pyrolysis fractions. The figure shows that in the conventional catalytic pyrolysis run most of the chlorine (more than 60 wt% of total original chlorine) goes as organochlorinated compounds to the liquid fraction, while around 35 wt% of total chlorine evolves as HCl to the gas fraction. This ratio is reversed by means of stepwise pyrolysis, which transfers up to 83.5 wt% of the chlorine to the gas fraction when the dechlorination step is carried out without catalyst prior to the catalytic pyrolysis step.

4. Conclusions

Catalytic pyrolysis is an attractive alternative for recycling mixed plastic waste. ZSM-5 zeolite is an effective catalyst which enhances the production of lighter and more aromatic liquid products and C3–C4 rich gas fractions from polyolefinic feedstocks. Pyrolysis products may be used as high HHV alternative fuels or as a source of valuable chemicals. However, the presence of PVC in the original sample leads to the generation of chlorinated chemicals which are part of the pyrolysis liquids, and this is very detrimental for their potential applications.

The liquids chlorine content can be reduced up to more than 75 wt% compared to conventional catalytic pyrolysis by means of a dechlorination step at 300 °C for 60 min. However, the authors

have proved that when the catalyst is mixed with the sample from the beginning of the process, the dechlorination step significantly decreases the catalyst activity, giving rise to liquids with a higher content of heavier hydrocarbons and a lower content of aromatics. On the contrary, the addition of the catalyst after the thermal dechlorination step has been found to be an efficient method to obtain almost the same results as in the conventional one-step catalytic process, in terms of products yields and quality, and at the same time achieve the high reduction in the chlorine content of the liquids.

To summarize, the main conclusion is that the optimum configuration for recycling plastic waste by pyrolysis whenever the income material contains PVC is a staged process consisting of a low temperature (300 °C) dechlorination step, complemented with an appropriate HCl capture and neutralization unit, and then a higher temperature catalytic step. The addition of catalyst should be done after the dechlorination step, since otherwise the catalyst is deactivated during such step.

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References

- [1] W.H. Starnes Jr., Structural and mechanistic aspects of the thermal degradation of poly(vinyl chloride), *Prog. Polym. Sci.* 27 (2002) 2133–2170.
- [2] Green Paper Environmental issues of PVC. European Commission 2000; COM (2000) 469. Available at: <http://ec.europa.eu/environment/waste/pvc/index.htm> (Last accessed: 20.10.11).
- [3] K. Wittstock, Plastics recovery-the end is the beginning. BASF Aktiengesellschaft, in: *Exporecicla Technical Conferences 2008*, Zaragoza, Spain, 2008.
- [4] Vinyloop® process. Innovation in PVC recycling. Available at: <http://www.vinyloop.com/> (Last accessed: 20.10.11).
- [5] A. López, I. de Marco, B.M. Caballero, M.F. Laresgoiti, A. Adrados, Pyrolysis of municipal plastic wastes: influence of raw material composition, *Waste Manag.* 30 (2010) 620–627.
- [6] A. Demirbas, Pyrolysis of municipal plastic wastes for recovery of gasoline-range hydrocarbons, *J. Anal. Appl. Pyrolysis* 72 (2004) 97–102.

- [7] J. Aguado, D.P. Serrano, J.M. Escola, Catalytic upgrading of plastic wastes, in: J. Scheirs, W. Kaminsky (Eds.), *Feedstock Recycling and Pyrolysis of Waste Plastics*, John Wiley & Sons, Chichester (UK), 2006, pp. 73–110.
- [8] A. López, I. de Marco, B.M. Caballero, M.F. Laresgoiti, A. Adrados, Influence of time and temperature on pyrolysis of plastic wastes in a semi-batch reactor, *Chem. Eng. J.* 173 (2011) 62–71.
- [9] A. López, I. de Marco, B.M. Caballero, M.F. Laresgoiti, A. Adrados, Dechlorination of fuels in pyrolysis of PVC containing plastic wastes, *Fuel Process. Technol.* 92 (2011) 253–260.
- [10] H. Bockhorn, A. Hornung, U. Hornung, Stepwise pyrolysis for raw material recovery from plastic waste, *J. Anal. Appl. Pyrolysis* 46 (1998) 1–13.
- [11] H. Bockhorn, A. Hornung, U. Hornung, P. Jakobströer, M. Kraus, Dehydrochlorination of plastic mixtures, *J. Anal. Appl. Pyrolysis* 49 (1999) 97–106.
- [12] A. Hornung, H. Bockhorn, J. Hentschel, U. Hornung, Environmental engineering: stepwise pyrolysis of plastic waste, *Chem. Eng. Sci.* 54 (1999) 3043–3051.
- [13] W. Kaminsky, J.S. Kim, Pyrolysis of mixed plastics into aromatics, *J. Anal. Appl. Pyrolysis* 51 (1999) 127–134.
- [14] A. Fontana, P.H. Laurent, C. Kestemont, C. Braekman-Danheux, Municipal waste pyrolysis (1): the behaviour of chlorine with cellulose and lignin, *Erdöl Erdgas Kohle* 116 (2000) 89–92.
- [15] M. Beckmann, A. Fontana, P.H. Laurent, C.G. Jung, J. Gehrman, Municipal waste pyrolysis (2): chlorine capture by addition of calcium and sodium-based sorbents, *Erdöl Erdgas Kohle* 117 (2001) 1–5.
- [16] A.G. Buekens, H. Huang, Catalytic plastics cracking for recovery of gasoline range hydrocarbons from municipal plastic wastes, *Resour. Conserv. Recycl.* 23 (1998) 163–181.
- [17] G. Elordi, M. Olazar, G. Lopez, M. Amutio, M. Artetxe, R. Aguado, J. Bilbao, Catalytic pyrolysis of HDPE in continuous mode over zeolite catalysts in a conical spouted bed, *J. Anal. Appl. Pyrolysis* 85 (2009) 345–351.
- [18] Y. Sakata, M.A. Uddin, A. Muto, Degradation of polyethylene and polypropylene into fuel oil by using solid acid and non-acid catalysts, *J. Anal. Appl. Pyrolysis* 51 (1999) 135–155.
- [19] Y.Z. Wang, C. Tang, Q. Zhou, L. Zheng, Catalytic effect of Al–Zn composite catalyst on the degradation of PVC-containing polymer mixtures into pyrolysis oil, *Polym. Degrad. Stab.* 81 (2003) 89–94.
- [20] J. Yanik, T. Karayildirim, M. Yuksel, M. Saglam, C. Vasile, H. Bockhorn, The effect of some fillers on PVC degradation, *J. Anal. Appl. Pyrolysis* 75 (2006) 112–119.
- [21] M. Blazsó, E. Jakab, Effect of metals, metal oxides, and carboxylates on the thermal decomposition processes of poly (vinyl chloride), *J. Anal. Appl. Pyrolysis* 49 (1999) 125–143.
- [22] Y. Sakata, T. Bhaskar, J. Kaneko, A. Muto, E. Jakab, T. Matsui, M.A. Uddin, Pyrolysis studies of PP/PE/PS/PVC/HIPS-Br plastics mixed with PET and dehalogenation (Br, Cl) of the liquid products, *J. Anal. Appl. Pyrolysis* 72 (2004) 27–33.
- [23] K. Murata, M. Brebu, Y. Sakata, The effect of PVC in thermal and catalytic degradation of polyethylene, polypropylene and polystyrene by a continuous flow reactor, *J. Anal. Appl. Pyrolysis* 86 (2009) 33–38.
- [24] Q. Zhou, W. Lan, A. Du, Y. Wang, J. Yang, Y. Wu, K. Yang, X. Wang, Lanthania promoted MgO: simultaneous highly efficient catalytic degradation and dehydrochlorination of polypropylene/polyvinyl chloride, *Appl. Catal. B* 80 (2008) 141–146.
- [25] J. Yanik, M.A. Uddin, K. Ikeuchi, Y. Sakata, The catalytic effect of Red Mud on the degradation of poly (vinyl chloride) containing polymer mixture into fuel oil, *Polym. Degrad. Stab.* 73 (2001) 335–346.
- [26] I. de Marco, B.M. Caballero, A. López, M.F. Laresgoiti, A. Torres, M.J. Chomon, Pyrolysis of the rejects of a waste packing separation and classification plant, *J. Anal. Appl. Pyrolysis* 85 (2009) 384–391.
- [27] A. López, I. de Marco, B.M. Caballero, M.F. Laresgoiti, A. Adrados, A. Aranzabal, Catalytic pyrolysis of plastic wastes with two different types of catalysts: ZSM-5 zeolite and Red Mud, *Appl. Catal. B* 104 (2011) 211–219.
- [28] M.F. Laresgoiti, B.M. Caballero, I. de Marco, A. Torres, M.A. Cabrero, M.J. Chomon, Characterization of the liquid products obtained in tyre pyrolysis, *J. Anal. Appl. Pyrolysis* 71 (2004) 907–934.
- [29] I. de Marco, B.M. Caballero, M.A. Cabrero, M.F. Laresgoiti, A. Torres, M.J. Chomon, Recycling of automobile shredder residues by means of pyrolysis, *J. Anal. Appl. Pyrolysis* 79 (2007) 403–408.
- [30] E.A. Williams, P.T. Williams, Analysis of products derived from the fast pyrolysis of plastic wastes, *J. Anal. Appl. Pyrolysis* 40–41 (1997) 347–363.
- [31] N. Grittner, W. Kaminsky, G. Obst, Fluid bed pyrolysis of anhydride-hardened epoxy resins and polyether–polyurethane by the Hamburg process, *J. Anal. Appl. Pyrolysis* 25 (1993) 293–299.
- [32] V. Vasile, H. Pakdel, B. Mihai, P. Onu, H. Darie, S. Ciocâlțeu, Thermal and catalytic decomposition of mixed plastics, *J. Anal. Appl. Pyrolysis* 57 (2001) 287–303.
- [33] D.P. Serrano, J. Aguado, J.M. Escola, J.M. Rodríguez, Influence of nanocrystalline HZSM-5 external surface on the catalytic cracking of polyolefins, *J. Anal. Appl. Pyrolysis* 74 (2005) 353–360.
- [34] A. Marcilla, M.I. Beltrán, R. Navarro, Thermal and catalytic pyrolysis of polyethylene over HZSM5 and HUSY zeolites in a batch reactor under dynamic conditions, *Appl. Catal. B* 86 (2009) 78–86.
- [35] D.P. Serrano, J. Aguado, J.M. Escola, Catalytic conversion of polystyrene over HMCM-41, HZSM-5 and amorphous SiO₂–Al₂O₃ comparison with thermal cracking, *Appl. Catal. B* 25 (2000) 181–189.
- [36] T.T. Wei, K.J. Wu, S.L. Lee, Y.H. Lin, Chemical recycling of post-consumer polymer waste over fluidizing cracking catalysts for producing chemicals and hydrocarbon fuels, *Resour. Conserv. Recycl.* 54 (2010) 952–961.
- [37] W.C. Huang, M.S. Huang, C.F. Huang, C.C. Chen, K.L. Ou, Thermochemical conversion of polymer wastes into hydrocarbon fuels over various fluidizing cracking catalysts, *Fuel* 89 (2010) 2305–2316.